

Preparations for ESO/SRC Conference Preparations are under way for the ESO I SAC conference on "Research Programmes for the New Large Telescopes", to be held at CERN on May 27-31, 1974. This conference will be in line with previous ESO meetings dealing with large telescope design and with auxiliary instrumentation developments, and the emphasis is to be on the southern hemisphere. With three large southern telescopes soon becoming operational (CTIO, Anglo-Australian and ESO), there are good reasons for having joint discussions with participants from the large projects outside ESO concerning the first areas for research with these instruments. The programme as presently outlined will comprise seven half-day sessions -one afternoon will be devoted to an excursion to the CERN laboratories -covering the following topics: Research programmes for large telescopes now in operation, Southern hemisphere problems, Observational cosmology, Instrumental capabilities, Philosophy of telescope use. The conference will start on Monday afternoon with an introductory talk given by Dr. J. L. Greenstein of the California Institute of Technology and the Hale Observatories, dealing with large telescope astronomy in general. Dr. Greenstein has been chairman of the Astronomy Survey Committee of the U. S. National Academy of Sciences, whose task it was to consider the need for major new astronomical facilities in the United States during the seventies. The results and recommendations of the Survey Committee are embodied in two volumes, entitled "Astronomy and Astrophysics for the 1970s". Clearly Dr. Greenstein's involvement in this project makes him eminently suited to give the introductory talk at the conference. Dr. Greenstein's lecture will be followed by shorter contributions by speakers from the big observatories -Hale, Lick and Kitt Peak -on research carried out or planned for the near future at these institutes. Following the first introductory session, lectures on specialized topics are scheduled for the next day: infrared astronomy, nearby galaxies, quasi-stellar objects, globular clusters in the southern sky and so on. In the afternoon session of the second day, theoretical aspects will be covered, an introductory paper being followed by shorter contributions on specialized subjects. The Wednesday morning session will be devoted entirely to the Magellanic Clouds, on which a special symposium was arranged in Santiago in March, 1969, in connection with the inauguration of the ESO Observatory on La Silla.

The afternoon is reserved for an excursion to the CERN facilities. The Thursday morning session will deal with observational cosmology, an introductory talk followed by contributions presenting optical and radio-astronomical aspects of cosmological problems. In the Thursday afternoon session instrumental capabilities will be discussed from a classical point of view. The new photographic emulsions and new powerful techniques for sensitizing photographic plates are bringing about a renaissance in astronomical photography. This session will be in the form of a panel discussion with four introductory speakers, dealing with the astronomical applications of the new emulsions, the extraction of information from photographic plates using fast computer technique and further reports on the ESO Sky Atlas and the parallel Science Research Council southern sky Schmidt project. The last session, on Friday morning, will be in the form of a panel discussion on a subject being raised probably for the first time at a conference: the philosophy of telescope use. This discussion concerning the allocation of time on large telescopes, with reference to the very high hourly operational costs, the search for more efficient methods of telescope use and so on, will undoubtedly give rise to a varied exchange of opinions.

The Big Telescope Building Shows Up at La Silla

From its inception in March, 1973, it took just nine months for the first visible bit of our building to appear over the rocky soil of La Silla's highest mountain top. Now it has only to rise and grow. For several months past the eruptions from rock-blasting on the platform of the truncated peak have provided a little spectacle on most afternoons. Since 1968 this platform has offered, as it were, a standing invitation to further activities. Now about one half of it has been levelled down two metres more in order to make sufficient room for the building. Another two metres down in the earth are trenches and holes blasted into the rocky soil. On the remaining half of the platform the contractor installed his site facilities, consisting of a carpenter's shop, a magazine and office barrack, and a concrete plant with storage for gravel and sand. In January, 1974, a full view could finally be obtained of the octagonal concrete foundation for the pier which will later support the telescope. Further out, a circular trench of 30 m diameter for the outside wall surrounds the light grey plate. It goes without saying that the first bucket filled with concrete and poured into its proper place was hailed as marking the

birth of the edifice, and this event was celebrated with a gay little ceremony.

ESO/Chile and CERN SB Division Make Agreement In January, the Technical Department of ESO/Chile concluded an agreement in principle with the CERN Sites and Buildings Division for a number of projects, including road improvement at Pelicano and La Silla, the central heating system, the low tension electrical distribution network and the water softening plant. CERN will carry out the studies for this work and propose solutions. A prime mover is H. Laporte, head of the SB Division, who has visited Chile in this connection, and M.-A. Peuch, Asst. Director, ESO/Chile, is also closely involved.

Exhibition on Role of ESO A historical picture exhibition on the role of ESO was held at CERN in the first half of November, 1973. This show is still available and can be placed at the disposal of any agency wishing to use it. It will probably be displayed at the Palais des Decouvertes in Paris, within the near future.

TP Division Extends Premises At the TP Division, Geneva, an assembly hall, mechanical workshop and optical laboratory are under construction and will be completed probably in July, 1974. The new buildings will be occupied mainly by W. Richter's group on the design and assembly of the 3.6 m telescope. The optical laboratory will be used by J. Van der Lans for tracking experiments and in cooperation with R. Wilson for optical experiments and tests. The MESSENGER is at present planned as a quarterly publication. Contributions for issue No. 2 should accordingly be received by the editor by July 15, at the latest. They may be sent directly to The Editor, The Messenger, ESO / Hamburg, or via the local correspondents, namely: Havlen, coordinator, Chile M. de Groot, La Silla (scientific matters) G. Anciaux, La Silla M. Becker, Santiago Rodgers, ESO TP Division, Geneva

3.6 m Telescope Headed for Testing and Erection The big telescope moves on towards its big day in 1976. In the Creusot-Loire plant at Saint-Chamond, near Grenoble, where a large part of the work has been done, all the major components have now been welded and stress released. Machining and the manufacture of all the smaller parts will be completed by the summer of 1974. Then, after testing of some 20 sub-assemblies, the assembly of the whole telescope will begin and it will be completed by January, 1975. A six-month period is allotted for the telescope tests and adjustment of the drives. The alignment procedures will also be determined during this time. In the latter half of 1975 the telescope will be

dismantled and packed for transport to Chile. All components, including the aluminizing plant, will go in the same ship. They will be unloaded at Huasco, a small port some 150 km north of La Silla. From there the convoy heads straight for the mountain. The firm of Creusot-Loire will then put the whole telescope together within about three months. After the mechanical tests, the mirrors will be aluminized and installed for the first time in the telescope. This event, initiating the final steps towards making the big telescope operational, is scheduled for 1976.

ESO/Hamburg Goes Over to Computer

At the Director-General's Office, the Finance Service switched over to computer on the first day of 1974. The firm selected to carry out the work was Treuarbeit AG Hamburg, and it used a Honeywell Bull 415 computer. The first tryout, in December, 1973, lasted one week and gave satisfactory results. An existing programme has been adjusted to the current requirements of ESO book-keeping; the future needs of the Organization are also taken into account. For the Hamburg Administration this means that the time-lag in operations is now reduced from one month to one week and there will be more time for non-routine work. However, the manual accounting system was continued until March 31 to permit regular checks on the computer output. Under the previous system, the ESO/Chile Administration received a single debit note giving the total figure and then made the break-down. This note is now replaced by a complete and detailed specification of expenditure in Hamburg, charged against their budget by budget item. Early in 1974, definite proposals were made concerning partial implementation of the ESO/Chile accounting in the EDP programme.

Wilson Completes Optics Study

ESO recently was able to make some return for the technical aid given by CERN when Dr. Raymond Wilson, towards the end of 1973, completed a study on the optics of scintillator counters for the 300 GeV experimental area. Dr. Wilson, who is British, took his doctorate at the Imperial College and came to ESO from Zeiss of Oberkochen in September, 1972. Acknowledged as one of the foremost experts in optical design, he has given many lectures on optics and instrumentation at international conferences, particularly at the last two ESO/CERN conferences.

ESO Pioneers with S-3000 Measuring Machine

An OPTRONICS S-3000 SPECSCAN

measuring machine has recently been installed at the ESO Sky Atlas Laboratory in Geneva where it is undergoing installation tests. This machine, which is the first

of its kind in Europe, is capable of measuring positions (to 1mi) and densities (to .02 D) of astronomical plates up to 14" x 14". It will first be used for positional calibration of the Sky Survey plates which are taken with the ESO and SAC Schmidt telescopes, and also for quality control by means of image evaluation. However, it is expected that the machine will attract users of Schmidt plates within ESO as well as astronomers from institutes in ESO countries who want to evaluate plate material they have received from ESO. The manufacturer is the Optronics Co. of Chelmsford, near Boston, USA.

New Electric Power Plant at La Silla

On March 7, news was received that all three diesel motor generator sets had just become operational. Within the following days they were taken fully into service for the supply of power to the Observatory. Begun in June, 1973, the new electric power plant is part of the general plan for developing the installations at La Silla to meet the energy requirements of the 3.6 m telescope. It was assembled by Motoren-Werke of Mannheim, Germany. The location is at Km 17.5 on the road from Pelicano to La Silla and about two kilometers from the hotel. The plant consists of a main engine-room measuring 22 x 8.5 m -i.e. 187 m -and 5.8 m high inside; moreover, a small room containing the 6,000 volt starting cells, the four cells for the transformers that raise the generator tension from 380 V to 6,000 V, and, finally, a small combined workshop-store. The plant will have three groups of diesel generators of 480 kVA each, for a start, with space reserved for a fourth group later on. Three external mazout tanks with a capacity of 150 m³ each will permit independent functioning for more than two months, with one group operating continuously at full strength. The operation will be entirely automatic and the characteristics of the generator groups have been determined with a view to permitting the Observatory site to be supplied normally by a single group, the second being held as a reserve or a complement, if need be, and the third for maintenance. The new power plant will eventually supply the whole La Silla site by means of an underground 6,000 V cable about two kilometers long. It will also supply the installations at Pelicano and the pumping stations along the road through the present 6,000 V aerial electric cable connecting Pelicano with La Silla. It will replace the plant operating at El Pelicano since 1967. This has been supplying the whole ESO area through the aerial cable, but, with its three old groups of 115 kVA diesel generators, it

can no longer ensure an adequate supply for the Observatory. Filming the 3.6 m Telescope The Rodgers-Pillet film unit at Geneva is keeping well up with construction work on the 3.6 m telescope. Footage recently taken at the REOSC plant at Ballainvillers and Creusot-Loire at Saint-Chamond, also in France, has included some larger sections of the telescope, such as the horseshoe and the fork. The film, 16 mm soundtrack and in colour, follows the progress of the telescope, its aim being to provide a visual documentary record of the whole project. The producer is N. Rodgers and cameraman-for-Europe B. Pillet. They will bring together the material from the various construction locations and edit it at the ESO TP Division in Geneva. The finished product will be distributed outside ESO also -e.g. to observatories and teaching institutions -but not commercially.